

Reader Report: Week 06

Probabilistic Invariance: VI-BIP and the Core Theorems

Daniel Hardesty Lewis (dl3645)

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Summary

Wu and Blei (2025) recast ICP as sparse Bayesian variable selection. A binary selector $z \in \{0, 1\}^p$ marks invariant features; the posterior $p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \hat{g}(Y_i^e \mid X_{ei}^z) / \hat{p}_e(Y_i^e \mid X_{ei}^z)$ up-weights selectors for which the pooled conditional $\hat{g}$ matches each local $\hat{p}_e$ —a probabilistic analog of Peters et al.’s accept/reject invariance test.

Two theorems govern the posterior. **Consistency (Thm 1):** if $\mu_{\min} > 0$ (positive KL between pooled and local conditionals on any non-invariant feature), the posterior concentrates on z^* . **Contraction (Thm 2):** mass on $\{z \neq z^*\}$ decays as $O(R e^{-\kappa n E \mu_{\min}})$, where $R = 2^p$, E is environments, and n is per-environment n . Key implications: (i) data and environments trade off in the exponent—helpful given my $K \approx 4$ regimes; (ii) $R = 2^p$ is only a prefactor, so the dimensionality curse is milder than expected.

VI-BIP fits $q(z) = \prod_j \text{Bern}(z_j; \pi_j)$ by ELBO maximization. The obstacle is differentiating through discrete z ; the **U2G estimator** relaxes the discrete draw to a Gaussian surrogate, enabling standard stochastic gradient ascent. On a 6,170-gene perturbation dataset, VI-BIP matches exact BIP while being orders of magnitude faster.

Critique & Contrast with the Optimization Approach

Fan et al. (2024) replace the combinatorial posterior with a minimax problem $\min_b \max_{w \in U(\gamma)} \sum_e w_e \hat{\mathbb{E}}[(Y^e - b^\top X^e)^2]$, where $U(\gamma)$ permits negative weights. NegDRO yields a point estimate $\hat{\beta}^\gamma$ with polynomial-time convergence but no uncertainty quantification. The key difference: BIP exposes *which* features remain ambiguous when environments are few; NegDRO gives a fast answer but not a sense of what additional data would resolve. For my ~ 4 -environment, ~ 50 -feature NIMBYism project, the multi-modal BIP posterior is directly informative about where more environment diversity is needed.

Questions

1. The contraction bound uses the population $\mu_{\min}$, not an estimate. Is there a pre-test analogous to classical ICP’s global invariance check that certifies $\mu_{\min} > 0$ before fitting BIP?
2. VI-BIP’s mean-field family ignores posterior correlations between z_j and z_k . For strongly correlated regulatory features (zoning category, parking requirements), does mean-field bias π_j toward 0.5? Would a structured variational family (e.g. Ising) recover sharper posteriors?

References

- Y. Fan, S. Li, and Z. Yang. Causal invariance learning via efficient nonconvex optimization. *arXiv preprint arXiv:2412.11850*, 2024. URL <https://arxiv.org/abs/2412.11850>.
- Y. Wu and D. M. Blei. Bayesian invariance modeling of multi-environment data. *arXiv preprint arXiv:2506.22675*, 2025. URL <https://arxiv.org/abs/2506.22675>.